National College of Ireland

Project Submission Sheet

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I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.

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Signature:	Utkarsh Arun Satpute
Date:	04 / 04 / 2025

PLEASE READ THE FOLLOWING INSTRUCTIONS:

1. Please attach a completed copy of this sheet to each project (including multiple copies).
2. Projects should be submitted to your Programme Coordinator.
3. **You must ensure that you retain a HARD COPY of ALL projects**, both for yourown reference and in case a project is lost or mislaid. It is not sufficient to keep a copy on computer. Please do not bind projects or place in covers unless specificallyrequested.
4. You must ensure that all projects are submitted to your Programme Coordinator onor before the required submission date. **Late submissions will incur penalties.**
5. All projects must be submitted and passed in order to successfully complete the year. **Any project/assignment not submitted will be marked as a fail.**

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AI Acknowledgement Supplement

Your Name/Student Number	Course	Date
Utkarsh Arun Satpute	MSc. Data Analytics	04 / 04 / 2025

This section is a supplement to the main assignment, to be used if AI was used in anycapacity in the creation of your assignment; if you have queries about how to do this, please contact your lecturer. For an example of how to fill these sections out, please click here.

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
NA	NA	NA

Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

[Insert Tool Name]	
[Insert Description of use] **NA**	
[Insert Sample prompt] **NA**	[Insert Sample response] NA

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.

Forecasting Air Freight Demand Using Time Series

1) Abstract

The air freight is an important industry which play a crucial role in global trade and logistics, where predicting the accurate demand is essential as it can be useful in resource planning and operational management. This project presents a forecasting of air freight demand using Artificial Neural Networks (ANN). We have used this technique from the established research paper “Machine learning for Air Transport Planning and Management” by Wild et al in (2021). For our project we have trained on historical fright data from Indian airports between 2017 to 2023. To evaluate the model performance, Mean Absolute Error and Root Mean Squared Error were used. The findings of this study will help air freight industry by providing accurate, data-driven insights to improve real-time decision-making and operation planning.

2) Background & Scope

Technology has not only used for humans to come closer for contact but has also help for the industries to make them easier for their trade. As we been moving forward with the time the world his becoming smaller. Products which were exotic are now delivered to our doorsteps within days. The credit goes for the air freight and air shipping. The market size of air cargo industry is more than 175$ billion, making it the powerhouse of global trade, especially in the e-commerce industry [1]. These industries and their products and mainly rely on air fright because if its quick deliveries. As the global trade has been increasing rapidly and supply chain becoming complex, the need to accurately forecast of air fright demand become vital for increasing operations managing cost and to smoothen the flow of logistics.

But the demand of air freight can be Influenced by multiple factors which we will be discussing. In this the first is the seasonality, war like situations, economics and any sudden emergency like the pandemics or the supply chain bottlenecks.

The project aims to build a data driven forecasting time series model using the historical data which we have. The focus is to understand trends and pattern in fright volumes and apply predictive model to get future demand. The outcomes of this study can be used to support airports , airplanes, and logistics companies in making there their decision better and optimize the operations.

3) Goals

The primary goal of this project is to forecast air freight demand. Forecasting enables airlines, airport authorities, and logistics providers to optimize capacity planning, reduce operational disruptions, and respond more efficiently to market changes.

In today's rapidly evolving logistics landscape—driven by e-commerce growth, supply chain disruptions, and volatile trade routes—the ability to anticipate freight volumes is essential. As noted by Maersk, shipment forecasting improves operational efficiency by reducing guesswork, securing better transport pricing, and ensuring on-time delivery to meet customer demand.

This project aims to:

- Analyse historical freight data to uncover trends and seasonality.
- Deliver actionable insights that allow stakeholders to plan cargo space, allocate fleet resources, and optimize delivery schedules.

By achieving these goals, the research will:

- Strengthen the resilience of air freight systems against demand fluctuations.
- Support better decision-making in global logistics and trade.

4) Background and Literature Survey

The forecasting in logistics is crucial, especially in today's time with the growing importance of the air freight industry, where demand frequently fluctuates due to seasonal cycles, geopolitical events, and economic conditions. Traditional statistical models like ARIMA and Exponential smoothing have been commonly applied for forecasting, but they often fail limited in their ability to capture nonlinear patterns, and seasonal trends patterns in a large dataset. These led to growing interest in machine learning and deep learning approaches.

Recent research studies have introduced machine learning and deep learning techniques to improve forecasting accuracy. The first paper, "*Time Series Forecasting with High Stakes*" by Garg and Shukla (2024) [2], combined statistical and deep learning models like Temporal Fusion Transformers (TFT). Their work showed better results in capturing nonlinear dependencies, but these models required significant computational resources and complex model building.

The second paper was published in 2021 by Graham Wild et al. [3], where they used machine learning algorithms such as SVM, Regression Trees, and Artificial Neural Networks (ANN) for air transportation planning. Their results showed that ANN outperformed linear models, although they required tuning and careful data preparation.

The third paper focused on revenue management and optimization in air cargo. The study, *"AI-CARGO: A Data-Driven Air-Cargo Revenue Management System"* by Rizzo et al. (2019) [4], aimed to predict the discrepancy between booked and actual cargo volumes, helping reduce offloading costs and improve operational efficiency.

Based on this literature review we have implemented Artificial Neural Network (ANN). Unlike TFT used by Garg & Shukla, Ann is less computationally intensive and is also more suitable for our historical dataset. ANN offers better nonlinear learning capability as compared to SVM and Regression trees used by Wild et al.

In summary, ANN was used for its proven success from previous literature and its ability to model complex trends. ANN serves as reliable and efficient forecasting tool.

5) Ethical Concerns

One of the key concerns is the data privacy and confidentiality. In project implementations, logistics data may include sensitive business or customer information. Organizations must ensure compliance with data protection regulations such as the General Data Protection Regulation (GDPR) to prevent misuse of data leakage or unauthorized access [5].

Another major concern is the potential for bias in model prediction. IN the training data is imbalanced like for example, containing more data from busy metro airports the model may Favor certain regions leading to unequal or unfair resource allocation.

Moreover, reliance on AL-system without human oversight can lead to potential risks. Forecasting models should serve as decision- support tools rather than fully autonomous systems. The world Economic Forum highlight that AI must be applied responsibly with human judgment and domain expertise guiding the final decisions [6].

Finally, AI must be monitored and updated continuously. As Freight patterns can change due to policy shifts, pandemics or trade route. If we didn't retrain our model, it can be risky becoming it as an outdated or misleading. Ethical Al practice includes model lifecycle management as given by IBM's AI Governance Framework [7]

6) Implementation of the technique

Dataset Description and Preparation:

We have used dataset which contain 3,104 rows and 6 columns. Which contains daily records of air freight from 2017 to 2023 across various Indian airports.

To understand the better overall trend and seasonal effect, we plotted the time – series of total freight over time. We have plotted a time series which is shown in figure 1: Time – series plot of total air freight volume from 2017 to 2023.

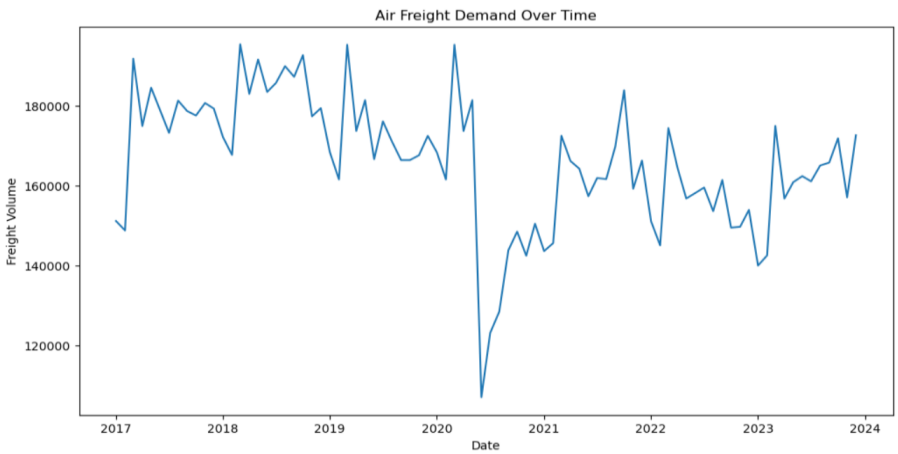

Figure 1: Air Freight Demand over time

From the above figure we can see that freight volumes fluctuate seasonally and significantly dipped at a certain period like a Covid – 19 pandemic which appeared in 2020, that's why we need a robust prediction model.

Feature Engineering

As we can see that from the above figure that time series is not a robust model. To enhance the prediction accurately we have some feature engineering which could result in recognizing temporal trends and seasonality in air freight demand.

First, we have extracted year, month and the day of the week from the date column which can capture both long-term and seasonal pattern. As this could happen when there are months of holidays or quarter- end shopping trends and may vary between the weekdays and weekends.

After that we have used lag feature i.e. (lag_1) these represent the volumes of freight from the previous days which result our model to learn from the recent activity and can capture short- term tends.

To stabilize the fluctuations and momentum of freight volume we have implemented the rolling mean average, and we have taken these for 3 days. This will smooth the noise out from daily variations and gives our model context about recent trends. These two features will help ANN to forecast the demand accurately.

Data Scaling and Model Configuration

We have divided our columns into two features which are X and Y and we have scaled them using StandardScaler which is important for Ann performance. Without the proper scaling the model might perform poorly.

Ann Architecture

We used ANN model and trained using Keras (TensorFlow Backend) with the following architecture. We had taken input layer has 5 and for first layer of hidden we used Dense (64 units, ReLU) + Dropout (0.2). The second layer has Dense of 32 units. The output layer has Dense 1 unit for regression output. For training parameters, we have used Adam has an optimizer and for calculating loss function we have used Mean squared error (MSE). For the Epochs we have taken 100 and Batch size is 16.

Model Evaluation

For training we split our data into 80% and for testing we will be doing on 20%. After assessing model performance, we used two standard metric Mean Absolute Error and Root Means Squared Error. The Ann model achieved MAE as a 6,510.72 and RMSE achieved as a 9,541.02. From this output we can state that the actual freight volume is ranging between the 150,000 to 200,000 units. Which indicate the actual prediction error between the 3 – 6% of actual values which could be acceptable in real world forecasting.

Visualization

To see our prediction more understandable we have visualized in figure 2: actual vs predicted air freight volumes on our dataset. From the prediction we can see that the values generated by ANN are close to the actual demand of freight over time. From these we can state that our model successfully learned the seasonal fluctuations and short-term variations.

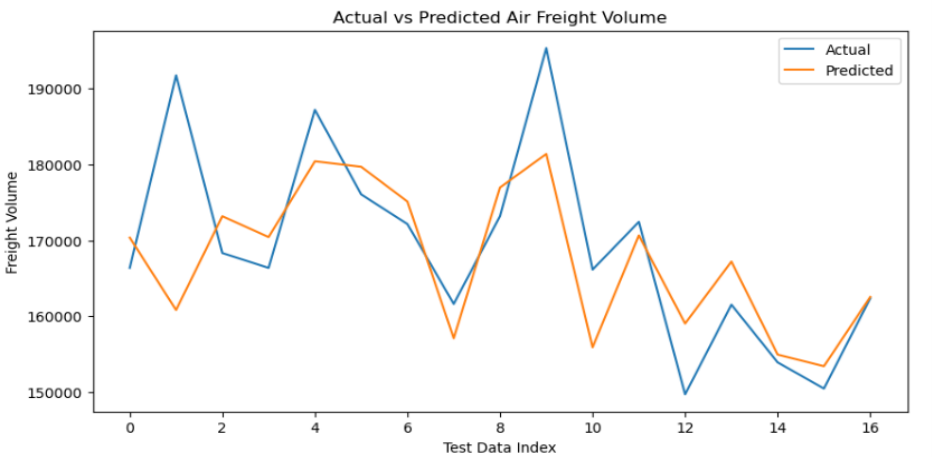

Figure2: Actual vs Predicted Air Freight Volumes Using ANN Model

The alignment of the two curves in capturing the movements, peaks and the dips from the figure state that our model's ability to generalize pattern. While minor deviations exist which is obvious while using a real-world forecasting, the Ann shows strong predicting behaviour especially in a period where there are gradual changes.

Summary

From the Wild et al. (2021) the implementation of Artificial Neural Network is reliable and accurate for forecasting the freight demand. The ability of model from learning the lag and seasonal changes resulted in a robust and reproducible forecasting pipeline.

7) The Quantitative Findings

The result of ANN was accepted by the two metric which are Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The MAE achieved 6,510.72 and RMSE achieved 9,541.02. Indicating low level of prediction error when compared to actual freight volume. As per this result we considered these has a highly acceptable
In forecasting applications involving logistics and transport planning.

Additionally, the actual vs predicted freight volume plot from figure 2 confirms that our model's ability to capture both the short-term fluctuations and seasonal patterns.

The Business values which our model brings for forecasting air fright demand with this level holds a significant strategic and operational value.

- Planning for Resource: Accurate demand forecasting will allow airport authorities to manage the logistics and help for airlines to optimize resource allocation like ground handling and store spacing eventually this will in reduce in cost and operational inefficiencies.
- Risk Mitigation: Understanding the upcoming demand pattern which will help for logistics companies prepare for high demand periods and can also adapt during low demand phase
- Strategic Forecasting: The insight will also help for company to plan for expansion of their infrastructure, manage the fleet and to optimize the route.

Overall, our model not only deliver reliable quantitative forecast but also empowers decision-makers in the air freight ecosystem to make data-driven, cost-efficient operational decisions.

8) Conclusion

The use of Artificial Neural Network to forecast air freight demand using the historical data was successfully demonstrated. The model we used was inspired by prior research and implemented using practical feature engineering, scaling and performance metrics. The model achieved low error margin which validates for using in real world applications. Moreover, the study highlights the importances of decision making and confirms that Ann is an efficient tool for demand prediction.

9) Future Work

While the ANN model performed well, future improvement could include the additional external features such as economic indicators, weather conditions and global trade activity for improving forecasting results.

10) References

[1] Maersk, "What is air freight: Meaning, benefits, and suitability," *www.maersk.com*, Apr. 04, 2024. Available: https://www.maersk.com/logistics-explained/transportation-and-freight/2024/04/04/choosing-air-freight

[2] Garg, A., & Shukla, N. (2024). Time Series Forecasting with High Stakes: A Field Study of the Air Cargo Industry. arXiv preprint. https://arxiv.org/abs/2407.20192

[3] Wild, G., Baxter, G., Srisaeng, P., & Richardson, S. (2021). Machine Learning for Air Transport Planning and Management. arXiv preprint. https://arxiv.org/abs/2112.01301

[4] Rizzo, S. G., Lucas, J., Kaoudi, Z., Quiane-Ruiz, J. A., & Chawla, S. (2019). AI-CARGO: A Data-Driven Air-Cargo Revenue Management System. arXiv preprint. https://arxiv.org/abs/1905.09130

[5] European Commission, "Data protection," *commission.europa.eu*, 2023. Available: https://commission.europa.eu/law/law-topic/data-protection_en

[6] A. Kaspersen and W. Wallach, "We're failing at the ethics of AI. Here's how we make real impact.," *World Economic Forum*, Jan. 14, 2022. Available: https://www.weforum.org/stories/2022/01/we-re-failing-at-the-ethics-of-ai-here-s-why/

[7] "The enterprise guide to AI governance," *IBM*, 2024. Available: https://www.ibm.com/thought-leadership/institute-business-value/en-us/report/ai-governance